Geometric Supervision for Road Segmentation

A Controlled, Pre-registered Study on Massachusetts Roads, DeepGlobe and Ottawa

Technical report (paper 2): data, full pipeline, architectures, targets, protocol, all results, error analysis, literature survey and comparison

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Abstract

We ask whether supervising a road-segmentation network with the *geometry* of the road — where its centre-line runs and how wide it is — improves road extraction from aerial and satellite images, and in which form. Starting from a strong ImageNet ResNeXt-50 U-Net (R0), we derive centre-line and width targets from every training mask by image processing (skeletonisation, distance transforms and a disc cover that reproduces the label *exactly*), and test, under one recipe, one data stream and pre-registered decision rules: (i) geometry as auxiliary training targets (GA); (ii) a mask *drawn* from predicted centre-lines and widths (GB); (iii) the two geometric terms separately (GAc, GAw); (iv) two prior auxiliary targets on the same model — Deep Distance Transform scale classes (GD) and road orientation (GO); and (v) geometry fed back into the decoder (GFa/GFb). Across 91 trained models on three datasets (55 new, ≈220 GPU-hours) we find one confirmed gain: GA improves per-image test IoU on DeepGlobe by +0.0073 (two-level bootstrap 95% CI $[+0.0033, +0.0118]$, seed-paired t $[+0.0011, +0.0135]$; pooled IoU 0.6771 → 0.6858) and beats DDT targets there (+0.0046, both intervals exclude zero). The gain is a reduction of false positives (−6 to −7%), needs both geometric terms, and does *not* replicate on Massachusetts (+0.0027, 5 seeds) or Ottawa (−0.0029, 5 seeds). Drawing the mask from the geometry (GB) and feeding the geometry back (GFa) bring no detectable improvement anywhere. Our baseline is level with published results under each dataset's protocol (e.g. Ottawa mIoU 0.934 vs. 0.934 for RoadNet on the same test scenes).

Contents

1 Summary of findings **3**

2 Question, motivation and history of the study **3**

3 Literature survey **4**
3.1 Road extraction as segmentation 4
3.2 Connectivity and topology 4
3.3 Geometric and auxiliary targets (the closest prior work) 4
3.4 Label noise in road datasets 5
3.5 Position of this study 5

4 Datasets **5**
4.1 Massachusetts Roads 7
4.2 DeepGlobe Road Extraction 7
4.3 Ottawa (RoadNet) 7

5 The pipeline, step by step **7**

6 Architectures **9**
6.1 Shared encoder–decoder (R0) . . . 9
6.2 GA: geometric auxiliary heads . . . 9
6.3 GB: the mask drawn from centre and width . . . 9
6.4 GF: geometry fed back into the decoder . . . 10
6.5 Baselines on the same model . . . 10

7 Geometric targets **11**

8 Loss functions **12**

9 Metrics **12**

10 Statistical analysis **12**
10.1 Pre-registration . . . 13

11 Results **14**
11.1 All metrics . . . 14
11.2 Pre-specified and secondary comparisons . . . 16
11.3 Attribution of GA's DeepGlobe gain . . . 16
11.4 Prior geometric targets . . . 17
11.5 Geometry feedback and its selection . . . 17
11.6 Error analysis . . . 17
11.7 Training curves . . . 18
11.8 Qualitative results . . . 19

12 Comparison with published results **20**

13 Discussion **20**

14 Limitations **21**

15 Reproducibility **21**

A Paper-1 models in the same table **22**

1 Summary of findings

Table 1: Headline results. $\hat{\Delta}$ = difference in per-image test IoU, mean over images of the seed-averaged difference. "Difference" requires both the two-level bootstrap and the seed-paired t 95% intervals to exclude zero with the same sign (Section 10). All tests were fixed in code before the runs they test were trained (or, for the DeepGlobe confirmatory test, before its result was read).

Question	Dataset	$\hat{\Delta}$	Decision	Status
Does drawing the mask from centre+width help? (GB − R0)	Massachusetts	−0.0012	no detected difference	primary
	DeepGlobe	−0.0017	no detected difference	primary
Does geometric auxiliary supervision help? (GA − R0)	Massachusetts	+0.0027	no detected difference	primary, 5 seeds
	DeepGlobe	**+0.0073**	**difference**	confirmatory
	Ottawa	−0.0029	no detected difference	primary, 5 seeds
Are our targets better than DDT's? (GA − GD)	DeepGlobe	**+0.0046**	**difference**	primary
Does feeding geometry back help? (GFa − R0)	Massachusetts	+0.0032	no detected difference	primary
	DeepGlobe	+0.0056	inconclusive	primary
	Ottawa	−0.0032	no detected difference	primary

1. **One confirmed, dataset-specific gain.** Auxiliary centre-line + width supervision (GA) improves DeepGlobe road IoU by about +0.9 points pooled; the effect is consistent across seeds $(+0.0070, +0.0049, +0.0099)$ and survives the strict two-interval rule. It does not replicate on Massachusetts or Ottawa.
2. **The gain is fewer false positives.** On DeepGlobe GA removes 6.1% of the false positives next to roads and 7.2% of those far from roads, while recall by road width is unchanged (Section 11.6).
3. **Both terms are needed; width does most.** Width-only (GAw) gets most of the pooled gain (0.6849 vs. 0.6858) but is not confirmed alone; centre-only (GAc) does nothing; GA beats both.
4. **Better than a prior geometric target on this pipeline.** GA beats Deep Distance Transform targets (GD) on DeepGlobe; neither GD nor orientation (GO) improves on R0 anywhere. The width encoding alone (GAw) is not distinguishable from DDT's (GAw − GD = +0.0016).
5. **Representations and feedback do not help.** Drawing the mask (GB), and feeding the predicted geometry into the decoder (GFa), give no detectable improvement over R0 or over GA.
6. **Most errors are at road edges.** 69% (Massachusetts), 51% (DeepGlobe) and 63% (Ottawa) of R0's error pixels lie within 2 px of a label edge — where labels are least certain.

2 Question, motivation and history of the study

Starting point. Paper 1 (separate report, `docs/report/roadseg_report.tex`) tested a wavelet (DWT) front-end on a ResNeXt-50 U-Net and found no effect. Its baseline R0 is a strong, current model (DeepGlobe pooled IoU 0.677). Paper 2 changes what the network is *taught*, not its input.

Motivation: the labels' geometry. Measured on training masks (Figure 2), Massachusetts labels are rasterised centre-lines of nearly constant width (5–7 px), while DeepGlobe and Ottawa labels have widths that vary by a factor of 5–8. A label is reliable about *where a road runs* and less so about *where its edges are*: most of the baseline's errors sit within 2 px of a label edge (Section 11.6). This suggested supervising the network with a representation of the road that separates the reliable part (centre-line) from the less reliable part (width), and measuring the effect on the standard metric, road IoU.

Sequence of decisions (all recorded in the ledger).

1. User's choices: direction = label geometry; headline metric = standard road IoU; approach = ladder R0 → GA (auxiliary) → GB (drawn mask).
2. Target design. A first version (skeleton + width from the distance transform) could only redraw 0.91 (Massachusetts) / 0.92 (DeepGlobe) of its own labels; diagnosis showed 1-px staircase edges on slanted roads. The adopted *disc cover* redraws every training label exactly (IoU 1.000, Section 7). Checked on training masks only.
3. Campaign `geo`: GA, GB on Massachusetts and DeepGlobe; primary GB − R0.
4. Massachusetts showed GA − R0 +0.0024 with a significant t interval only; GA − R0 on DeepGlobe was registered as a confirmatory test *before* its result was read, and confirmed.
5. Campaign `geo2`: seeds 3–4 for R0/GA on Massachusetts (the decision rule's follow-up) and the attribution GAc/GAw on DeepGlobe.
6. Literature search: Deep Distance Transform [23] is the closest prior method. Campaign `geo3`: DDT targets (GD) and orientation (GO) on GA's model; primary GA − GD.
7. Third dataset: Ottawa (RoadNet [25]); CHN6-CUG [33] needs a request form and was not used. Campaign `ott`: R0, GA, GD × 5 seeds.
8. User asked for an architectural addition. Guardrails agreed in advance: select on validation only, pre-register, report every variant. Campaigns `gfsel` (two feedback variants, seed 0, selection on validation) and `gfconf_a` (the selected variant).

3 Literature survey

3.1 Road extraction as segmentation

Mnih and Hinton [1] introduced patch-based road labelling and the Massachusetts Roads benchmark with relaxed precision/recall [2]. Fully convolutional networks [3], U-Net [4] and the residual U-Net [5] followed. On DeepGlobe [6], D-LinkNet [7] (ResNet-34 LinkNet with a dilated centre block) won the 2018 challenge; NL-LinkNet [8] replaced the dilated block with non-local operations. Transformer and hybrid encoders dominate recent reports: SegFormer [10], Seg-Road [11] (transformer encoder, CNN decoder and a pixel-connectivity head), URoadNet [12], DSFC-Net [13], RoadFocusNet [14].

3.2 Connectivity and topology

Pixel losses tolerate small gaps that break a network. DeepRoadMapper [15] reasons over a graph; Mosinska et al. [16] add a perceptual topology loss; clDice [17] optimises soft-skeleton overlap; CoANet [18] uses strip convolutions and connectivity attention; Seg-Road's PCS [11] predicts 8-neighbour connectivity. Connectivity is evaluated with APLS [19]. Graph-output methods (RoadTracer [20], Sat2Graph [21], SAM-Road [22]) are judged on graph metrics.

3.3 Geometric and auxiliary targets (the closest prior work)

- **Deep Distance Transform (DDT)** [23], for tubular structures in CT: a segmentation head plus a head that classifies the distance transform, rounded into K "scale classes", at *every* foreground voxel (classification because regression is unstable under annotation error); at inference a shape is reconstructed from a pseudo-skeleton and the predicted scales (balls), refining the mask. Our GA is DDT's multi-task part with different targets; GB is a trained, differentiable form of its

reconstruction; GD reproduces its targets on our model. We found no application of DDT to roads.

- **Orientation** [24]: road orientation (binned angles) learned jointly with segmentation in stacked multi-branch hourglasses, plus a connectivity refinement stage; reported +7.5% APLS on DeepGlobe. GO uses orientation targets on our model.
- **Multi-task surface/edge/centre-line**: RoadNet [25] (the Ottawa dataset's paper) learns road surface, edges and centre-lines in cascade; a cascaded multi-task network [26] does the same for surface, centre-line and edge.
- **Centre-line + width**: Wei et al. [27] regress a centre-line confidence map and road width to extract vector centre-lines and edge lines (not to improve the mask).
- **Distance maps**: BrightEarth [28] uses segmentation, orientation and distance-map heads.
- **Other auxiliary tasks**: road-direction connectivity [29] and road location prediction [30].

3.4 Label noise in road datasets

Road masks derived from map centre-lines carry offset and width errors; noise-aware losses were proposed for topological label noise [31]. DeepGlobe is described as annotated with varying road widths, unlike fixed-width OSM-derived sets [32].

3.5 Position of this study

Most cited works report single runs without variance, with re-implemented baselines that often score far below their original papers (e.g. D-LinkNet on DeepGlobe: 0.6466 in [7], 0.5762 in [11], 0.6655 in our runs). We hold the model, recipe, data stream and seeds fixed, vary only the supervision, use 3–5 seeds with intervals that include seed variance, and fix every test in code before its runs are trained.

4 Datasets

Table 2: Datasets as used. Label widths are measured at skeleton pixels (2 EDT -1) on 40 training masks (Ottawa: all 12). *Ottawa is used at 0.42 m (exact 2× downsampling of 0.21 m).

Dataset	GSD (m)	train / val / test	example test tile (px)	label width p5 / p50 / p95 (px)
Massachusetts	1.0	1108 / 14 / 49	1500×1500	5.0 / 5.3 / 7.0
DeepGlobe	0.5	4358 / 933 / 935	1024×1024	4.7 / 9.0 / 19.4
Ottawa	0.42*	12 / 2 / 6	1792×2048	6.2 / 16.1 / 28.1



Figure 1: Training-set examples, two per dataset, each the 512 × 512 window with the most road pixels (a fixed rule, not a hand pick); right of each image, the label overlaid in orange. Massachusetts: fixed-width centreline labels at 1 m. DeepGlobe: rural and peri-urban scenes at 0.5 m, widths varying with road type. Ottawa: manually drawn road surfaces, including carriageways and parking aisles.

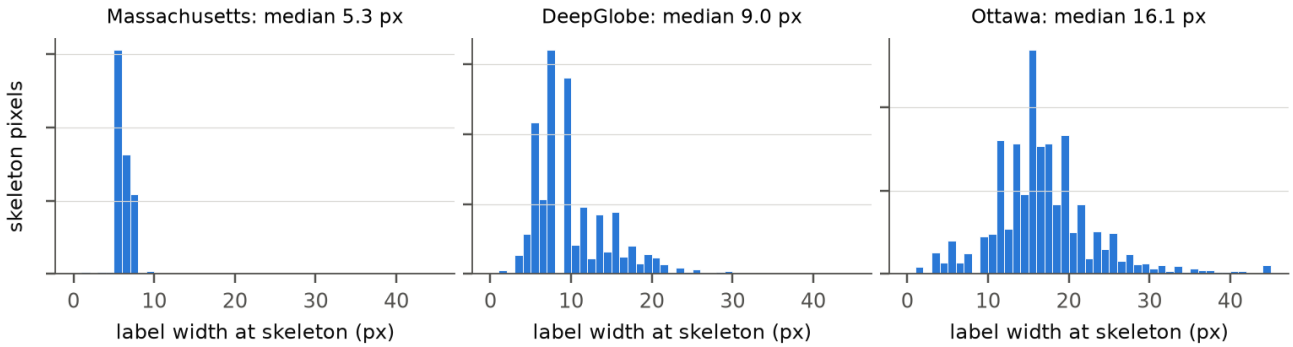


Figure 2: Distribution of label width at skeleton pixels (training masks). Massachusetts: nearly constant; DeepGlobe and Ottawa: broad.

4.1 Massachusetts Roads

Aerial RGB, 1 m/px, 1500 × 1500 tiles, labels rasterised from OpenStreetMap centre-lines [2]; official split 1108 / 14 / 49. **No-data masking**: a pixel is no-data if all channels are ≥ 250 and it lies in a 4-connected white region of ≥ 2500 px (11% of pixels on average, up to 79% in some tiles, with roads drawn through them). The validity mask is applied to the loss, crop sampling, selection and every metric, so Massachusetts numbers are not directly comparable with published ones that count those pixels.

4.2 DeepGlobe Road Extraction

Satellite RGB, 0.5 m/px, 1024 × 1024, Thailand/Indonesia/India [6]. Only the 6226 challenge training tiles have labels; they are re-split by image (seed 1337) into 4358 / 933 / 935. Test split fingerprint (SHA-256 of sorted ids): `a18e6da5...fe`. Source: HuggingFace mirror `lammtfkday/deepglobe-roads` (counts match; pixel identity with the Kaggle release not verified).

4.3 Ottawa (RoadNet)

Google Earth imagery of Ottawa at 0.21 m/px with manually annotated road surfaces, edges and centre-lines [25]; 20 scenes (the repository says 21). Download: authors' Google Drive zip (SHA-256 `aacaa827...73fb6`). Preparation (`scripts/prepare_ottawa.py`):

1. Image: exact 2 × 2 box average → 0.42 m. Mask: road = green channel < 128 (red labels incl. anti-aliased pixels), then 2 × 2 majority. Widths become 7–27 px (native 14–54 px exceed our largest width class); the smallest scene becomes 512 × 640.
2. Split: the authors' test scenes 1, 16–20; validation = scenes 9 and 14 (`default_rng(1337)` from the authors' training scenes 2–15); training = the other 12.
3. No-imagery strips at the right edge of scenes 2 and 1 are cropped (168 and 46 columns); test scene 1 keeps 1.45% black pixels (strip not a clean block). Every model is scored on the same pixels.
4. Training length: 64 crops per scene per epoch (48 steps), 200 epochs (9600 steps).

5 The pipeline, step by step

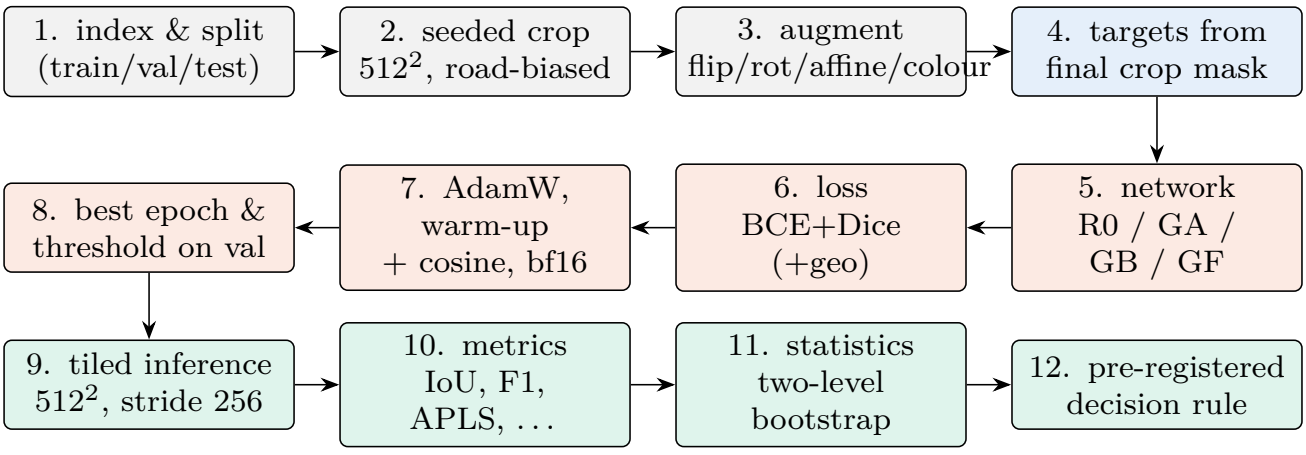


Figure 3: End-to-end pipeline. Grey: data; blue: geometric targets; orange: model and training; green: evaluation and inference. Steps 1–8 never open the test split.

1. **Index and split.** `data/prepare.py` builds the image/mask index per dataset and split (Section 4); a missing mask is an error, never silently skipped. Training records the split sizes and, for DeepGlobe, the test fingerprint.
2. **Seeded crops.** Each epoch draws $k\times$(number of training images) random 512×512 crops ($k = 8$; Ottawa $k = 64$). With probability 0.75 a crop is resampled (up to 12 tries) until it contains road on valid pixels; crops with $< 50\%$ valid pixels are rejected. Every crop is a pure function of (run seed, sample counter), so all models trained with one seed see the identical crop stream, also across preemption and resumption.
3. **Augmentation.** Horizontal/vertical flips and 90° rotations; with probability 0.5 an affine warp (rotation $\pm15°$, scale 0.9–1.1; bilinear image, nearest masks, reflective borders); with probability 0.5 brightness/contrast jitter in $[0.9, 1.1]$; ImageNet normalisation last.
4. **Geometric targets** (GA/GB/GAc/GAw/GF: disc cover; GD: DDT classes; GO: orientation) are computed from the *final, augmented* crop mask in the data-loader workers (44–49 ms per crop). They draw no randomness, so the crop stream is R0's.
5. **Network forward** (Section 6) under bf16 autocast; output = mask logits at input resolution (asserted), plus geometric logits for the geometric models.
6. **Loss** in fp32 (Section 8).
7. **Optimisation.** AdamW, $\eta_0 = 3\times10^{-4}$, weight decay 10^{-4}, batch 16, gradient clipping at 1.0 (a non-finite loss or gradient stops the run); per-iteration schedule: linear warm-up from $0.01\eta_0$ over 2 epochs, then cosine to $0.01\eta_0$. Massachusetts 100 epochs $\times$ 554 steps; DeepGlobe 40 $\times$ 2179; Ottawa 200 $\times$ 48. One recipe for every model.
8. **Model and threshold selection on validation.** After each epoch, pooled validation IoU at threshold 0.5 on non-overlapping tiles; best epoch kept. After training, the threshold $\tau^* \in \{0.20, 0.25, \ldots, 0.80\}$ maximising pooled validation IoU is frozen.
9. **Tiled inference.** 512×512 tiles at stride 256, last row/column aligned to the image edge; overlapping tile probabilities averaged; $\hat{Y} = \{P \geq \tau^*\}$. The test split is scored exactly once per run (a second scoring is refused); probability maps are saved as 8-bit PNG.
10. **Metrics** on stitched full-resolution images, valid pixels only (Section 9).
11. **Statistics**: per-image paired differences, two-level bootstrap and seed-paired t (Section 10).
12. **Decision** by the rule written before the runs (Section 10.1).

Experiment infrastructure. Every campaign is launched by `scripts/launch.sh`, which refuses a dirty tree and requires that the CPU rehearsal passed at the same commit; it freezes the code into `releases/<sha>/` and submits a SLURM array whose tasks import only that snapshot. Training checkpoints atomically every 5 min and on SIGTERM/USR1, resumes exactly (including the LR schedule and crop stream), and refuses to resume under different code. Each GPU job first runs a 2-step self-test (preflight). A CPU rehearsal trains every model in miniature on real data, scores, re-scores (must refuse), aggregates and runs two preemption drills. Results are aggregated by a dependent job. The test suite has 528 tests, all passing at the final commit.

6 Architectures

6.1 Shared encoder–decoder (R0)

Encoder: ResNeXt-50 32×4d [9], ImageNet weights; each bottleneck computes $y = x + \sum_{i=1}^{32} \mathcal{T}_i(x)$ (1×1 reduce, grouped 3×3 with 32 groups, 1×1 expand). For a 512^2 input it emits f_0 (stem, 64×256^2), f_1 (256×128^2), f_2 (512×64^2), f_3 (1024×32^2), f_4 (2048×16^2). **Decoder**: five stages $d_\ell = \phi_\ell([\mathrm{Up}_\ell(d_{\ell-1}); f_{4-\ell}])$, $d_0 = f_4$, with Up = bilinear $\times 2$ + 3×3 conv, ϕ_ℓ = two Conv–BN–ReLU, widths 512, 256, 128, 64, 32 at $32^2 \ldots 512^2$; no skip at full resolution. **Head**: 1×1 conv $\to$ one logit per pixel.

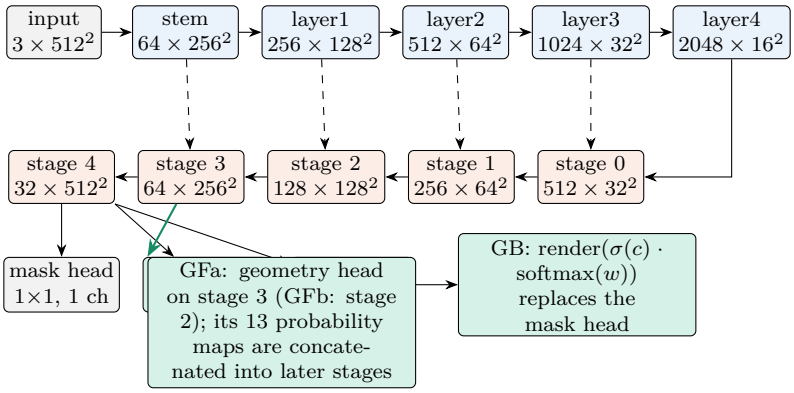


Figure 4: Shared ResNeXt-50 U-Net (R0) and the geometric variants. Dashed: skip connections. GA adds the two green heads on the 32-channel full-resolution features as auxiliary outputs (discarded at inference); GB draws the mask from them; GF moves the geometry head into the decoder and feeds its output back.

6.2 GA: geometric auxiliary heads

Two 1×1 convolutions on the final 32-channel feature map: a centre-line logit c and 12 width-class logits w (radii $\mathcal{R} = \{0, 1, 2, 3, 4, 5, 6, 8, 10, 13, 16, 20\}$ px). The mask comes from R0's own head; the geometric heads only add training signal and are not used at inference (+429 parameters, identical inference cost). Built after R0's modules, so at a given seed every shared weight is bitwise identical to R0's.

6.3 GB: the mask drawn from centre and width

GB uses the same heads but its mask is $m = \mathrm{render}(\sigma(c) \odot \mathrm{softmax}(w))$: each candidate centre pixel is dilated by its radius class, with an octagonal disc that is the Minkowski sum of alternating 3×3 square/cross steps.

Algorithm 1 `render`: variable-radius dilation in one pass of max-pools

Require: contributions $q_k = \sigma(c)\,\mathrm{softmax}(w)_k$ for radii $r_k \in \mathcal{R}$

$o \leftarrow$ none
for $j = \max \mathcal{R}$ **down to** 0 **do**
  for each k with $r_k = j$ **do**
    $o \leftarrow \max(o, q_k)$
  end for
  if $j > 0$ **then**
    $o \leftarrow \mathrm{maxpool}_{3\times3}(o)$ if j odd, else cross-shaped max-pool
  end if
end for
return o ▷ each bin dilated by exactly its radius; differentiable; 20 max-pools

The logit $\log m - \log(1-m)$ (with m clamped to $[10^{-4}, 1-10^{-4}]$) is returned so thresholding, inference and scoring are R0's code unchanged. R0's mask head is kept but unused.

6.4 GF: geometry fed back into the decoder

A 1×1 geometry head (13 channels: 1 centre + 12 width) is applied after decoder stage 3 (64×256^2, GFa) or stage 2 (128×128^2, GFb). Its probabilities $[\sigma(c); \mathrm{softmax}(w)]$ are bilinearly resized and concatenated into the input of every later stage, whose first convolution gains 13 input channels. The geometric logits are upsampled to full resolution for the loss. Tests confirm that changing only the geometry head changes the mask, and that the mask loss alone trains the geometry head. Parameters: +4,589 (GFa), +12,909 (GFb).

6.5 Baselines on the same model

- **GAc / GAw**: GA with the width (GAc) or the centre (GAw) loss weight set to 0.
- **GD**: Deep Distance Transform targets [23]: at every road pixel, round(EDT) as one of 20 classes; no centre term (+693 parameters).
- **GO**: orientation [24]: the undirected direction of the nearest skeleton segment (structure tensor of the blurred skeleton) in 18 bins of 10° plus a background class, supervised at every pixel (+660 parameters).
- **Paper-1 models** (appendix): R1–R3 (wavelets), U-Net from scratch, D-LinkNet.

Table 3: Trainable parameters.

Model	Parameters	vs. R0
R0	46,503,137	—
GA, GB, GAc, GAw	46,503,566	+429
GD	46,503,830	+693
GO	46,503,797	+660
GFa	46,507,726	+4,589
GFb	46,516,046	+12,909
U-Net (scratch) / D-LinkNet	31,037,633 / 31,095,137	—

7 Geometric targets

Let M be the binary mask of a training crop.

1. **Padding.** M is padded by edge replication (21 px) so a road leaving the crop is not treated as ending there; results are cropped back.
2. **Disc depth.** For radius r, the disc D_r is the Minkowski sum of the steps $j = r, \ldots, 1$ (square for odd j, cross for even j) — exactly the shape `render` draws. With $E_0 = M$ and $E_j = E_{j-1} \ominus \text{step}(j)$, $E_r = M \ominus D_r$; the depth of a pixel is the largest $r \in \mathcal{R}$ with the pixel in E_r.
3. **Centre pixels** S: skeletonize(M), then every mask pixel not covered by $\bigcup_{p \in S}(p + D_{\text{depth}(p)})$ is added with its own depth (after cropping).
4. **Targets**: centre map $= \mathbf{1}_S$; width class = depth at S, ignored elsewhere and on no-data pixels.

Proposition 1 (Exactness)**.** *Every disc lies inside M (by the definition of depth) and every pixel of M is covered (by step 3), so* render *of the one-hot targets equals M.*

Measured on 60 training images per dataset: drawn-back IoU 1.000000 (worst image 1.000000) on full images and 512^2 crops. The skeleton alone would reach 0.8817 (Massachusetts) / 0.9177 (DeepGlobe); about half the centre pixels are skeleton, the rest are 1-px staircase edge pixels with radius 0–1. Design history (training masks only): skeleton + nearest radius to EDT $-o$ reached 0.806 / 0.910 / 0.881 for $o = 0.5/1.0/1.5$ (Massachusetts); adding equal-depth neighbours for even widths did not help (0.910); diagnosis showed every error to be a 1-px edge pixel on slanted roads.

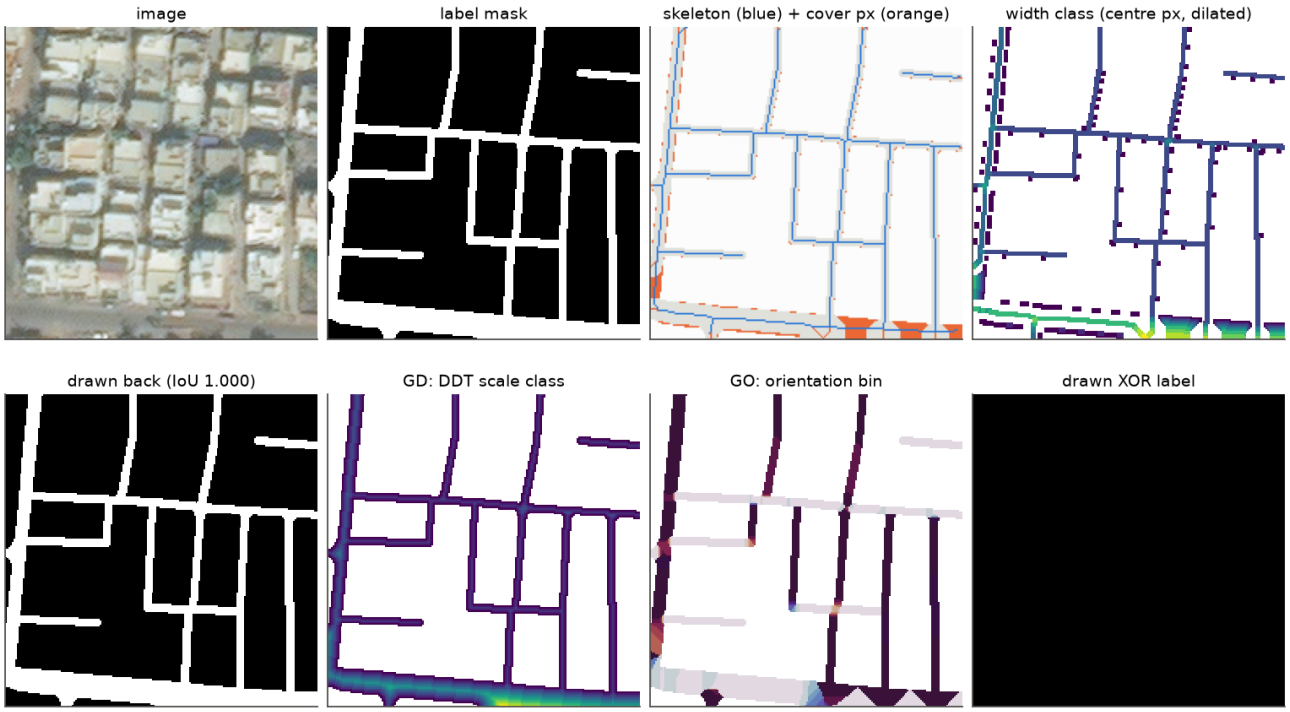


Figure 5: Targets on a DeepGlobe training crop (the 192^2 window with the most skeleton junctions). Top: image, label, skeleton (blue) and added cover pixels (orange), width class at centre pixels. Bottom: the label drawn back from the targets (exact), DDT scale classes (GD), orientation bins (GO), and the XOR of drawn-back and label (empty).

8 Loss functions

With logits z, $p = \sigma(z)$, label y and validity v:

$$\mathcal{L}_{\mathrm{BCE}}(z, y) = \frac{\sum v \,\mathrm{bce}(z, y)}{\max(\sum v, 1)}, \qquad \mathcal{L}_{\mathrm{Dice}}(z, y) = 1 - \frac{2 \sum p\, y\, v + 1}{\sum p\, v + \sum y\, v + 1}, \tag{1}$$

$$\mathcal{L}_{\mathrm{R0}} = \mathcal{L}_{\mathrm{BCE}}(z_m, y) + \mathcal{L}_{\mathrm{Dice}}(z_m, y), \tag{2}$$

$$\mathcal{L}_{\mathrm{geo}} = \mathcal{L}_{\mathrm{R0}} + \lambda_c[\mathcal{L}_{\mathrm{BCE}}(z_c, \mathbf{1}_S) + \mathcal{L}_{\mathrm{Dice}}(z_c, \mathbf{1}_S)] + \lambda_w \frac{1}{|S_v|} \sum_{u \in S_v} \mathrm{CE}(z_w(u),\, k(u)), \tag{3}$$

with $\lambda_c = \lambda_w = 1$ (untuned) for GA/GB/GF, $(1, 0)$ for GAc, $(0, 1)$ for GAw; for GD and GO the class term runs over road pixels (GD) or all valid pixels (GO) and $\lambda_c = 0$. For GB, z_m is the logit of the drawn mask. All losses in fp32; Dice smoothing 1 keeps road-free crops stable.

9 Metrics

Counts TP, FP, FN, TN over valid pixels of stitched test images at τ^*.

- **IoU** (road) $= \frac{TP}{TP+FP+FN}$, pooled (counts summed over images) and per image (IoU_{img}, the unit of the statistics).
- **mIoU** $= \frac{1}{2}(\frac{TP}{TP+FP+FN} + \frac{TN}{TN+FP+FN})$.
- **F1** $= \frac{2TP}{2TP+FP+FN}$, **precision** $= \frac{TP}{TP+FP}$, **recall** $= \frac{TP}{TP+FN}$, **accuracy** $= \frac{TP+TN}{TP+TN+FP+FN}$ (pooled).
- **Relaxed F1** ($\rho = 3$ px) [1], **clDice** [17], **APLS** [19] (graphs from skeletons, 500 node pairs per direction, 8 px snapping, a missing node scores 0), all per image then averaged.

Accuracy is dominated by background (roads are 4–15% of pixels) and is reported for completeness.

10 Statistical analysis

For arms $a \in \{0, 1\}$, seeds s and images $i = 1..N$, $\hat{\Delta} = \frac{1}{N}\sum_i(\bar{u}_{1,i} - \bar{u}_{0,i})$. **Two-level bootstrap** ($B = 10{,}000$, seed 0): resample seeds within each arm, then images (paired), percentile interval. **Seed-paired** t: $d_s = \frac{1}{N}\sum_i(u_{1,s,i} - u_{0,s,i})$ over common seeds, $\bar{d} \pm t_{0.975,n-1} s_d/\sqrt{n}$. **Decision rule**: *difference* if both intervals exclude zero with the same sign; *inconclusive* if exactly one does; *no detected difference* if neither. With 3 seeds the bootstrap can be anti-conservative (simulated coverage 0.86–0.98), hence the second interval. Secondary analyses carry no multiplicity correction and are labelled as such.

10.1 Pre-registration

Table 4: Analysis plans (`src/roadseg/stats.py`) and when they were fixed.

Plan	Primary	Fixed	Commit
`geo`	GB − R0 (each dataset)	before any GA/GB run	ffe5351
(ledger)	GA − R0, DeepGlobe	2026-10-15 13:29, before the DeepGlobe result was read	e48f87c
`geo2`	GA − R0, Massachusetts, 5 seeds	before any geo2 run	a6cc48b
`geo3`	GA − GD, DeepGlobe	before any GD/GO run	ba1e8e9
`ott`	GA − R0, Ottawa, 5 seeds	before any Ottawa run	8cb06c4
`gf_a/b`	GF − R0 (each dataset); variant chosen on validation	before any GF run	c430651

11 Results

11.1 All metrics

Table 5: Test metrics, mean over n seeds; best per dataset in bold. Pooled metrics except IoU_{img}, $\text{F1}_{\rho=3}$, clDice and APLS (per-image means). †Paper-1 models. GFb is the single selection run (seed 0), shown for completeness.

Model	n	IoU	mIoU	F1	Prec.	Rec.	Acc.	IoU_{img}	$\text{F1}_{\rho=3}$	clDice	APLS
Massachusetts											
R0 (ResNeXt-50 U-Net)	5	0.6556	0.8177	0.7920	0.7979	0.7862	0.9806	0.6519	0.9065	0.8804	0.8007
R1 (DWT input)†	3	0.6554	0.8176	0.7918	0.7967	0.7871	0.9805	0.6533	0.9087	0.8821	0.8029
R2 (wavelet pool)†	3	0.6545	0.8170	0.7912	0.7900	0.7924	0.9803	0.6520	0.9079	0.8822	0.8029
R3 (R2 + clDice)†	3	0.6553	0.8175	0.7918	0.7898	0.7937	0.9804	0.6511	0.9060	0.8828	**0.8040**
U-Net (scratch)†	3	**0.6632**	**0.8217**	**0.7975**	0.7961	**0.7989**	**0.9809**	**0.6624**	**0.9155**	**0.8895**	0.8006
D-LinkNet†	3	0.6515	0.8154	0.7890	0.7895	0.7884	0.9802	0.6486	0.9080	0.8806	0.8002
GA (aux. centre+width)	5	0.6588	0.8194	0.7943	0.8031	0.7856	0.9809	0.6546	0.9077	0.8822	0.8029
GB (drawn mask)	3	0.6555	0.8176	0.7919	0.7929	0.7909	0.9805	0.6508	0.9054	0.8831	0.7990
GFa (geometry feedback)	3	0.6591	0.8196	0.7945	**0.8051**	0.7842	**0.9809**	0.6552	0.9071	0.8832	0.8008
GFb (feedback, 1/4)	1	0.6584	0.8192	0.7940	0.8022	0.7859	0.9808	0.6548	0.9072	0.8839	0.8015
DeepGlobe											
R0 (ResNeXt-50 U-Net)	3	0.6771	0.8306	0.8074	0.7976	0.8175	0.9846	0.6751	0.8978	0.8665	0.7755
R1 (DWT input)†	3	0.6785	0.8314	0.8085	0.8005	0.8167	0.9847	0.6768	0.8988	0.8670	0.7777
R2 (wavelet pool)†	3	0.6823	0.8333	0.8111	0.8008	0.8217	0.9848	0.6790	0.8996	**0.8687**	**0.7814**
R3 (R2 + clDice)†	3	0.6743	0.8291	0.8054	0.7959	0.8153	0.9844	0.6723	0.8952	0.8683	0.7763
U-Net (scratch)†	3	0.6461	0.8141	0.7850	0.7745	0.7958	0.9827	0.6419	0.8764	0.8378	0.7193
D-LinkNet†	3	0.6655	0.8245	0.7992	0.7932	0.8053	0.9840	0.6669	0.8955	0.8613	0.7614
GA (aux. centre+width)	3	**0.6858**	**0.8352**	**0.8136**	**0.8087**	0.8186	**0.9851**	0.6824	0.8985	0.8659	0.7753
GB (drawn mask)	3	0.6781	0.8311	0.8082	0.7944	**0.8225**	0.9845	0.6733	0.8940	0.8680	0.7588
GAc (centre only)	3	0.6784	0.8313	0.8084	0.8018	0.8150	0.9847	0.6755	0.8976	0.8661	0.7759
GAw (width only)	3	0.6849	0.8347	0.8130	0.8069	0.8193	0.9851	0.6793	0.8962	0.8639	0.7748
GD (DDT targets)	3	0.6834	0.8339	0.8119	0.8055	0.8184	0.9850	0.6777	0.8954	0.8615	0.7717
GO (orientation)	3	0.6812	0.8328	0.8104	0.8039	0.8170	0.9849	0.6764	0.8982	0.8667	0.7768
GFa (geometry feedback)	3	0.6850	0.8348	0.8131	0.8077	0.8185	0.9851	0.6806	0.8968	0.8651	0.7745
GFb (feedback, 1/4)	1	0.6857	0.8352	0.8136	0.8085	0.8187	**0.9851**	**0.6830**	**0.9002**	0.8682	0.7805
Ottawa											
R0 (ResNeXt-50 U-Net)	5	**0.8861**	**0.9342**	**0.9396**	0.9327	**0.9466**	**0.9843**	**0.8798**	**0.9783**	**0.9609**	**0.8828**
GA (aux. centre+width)	5	0.8828	0.9322	0.9377	0.9359	0.9396	0.9839	0.8769	0.9760	0.9542	0.8394
GD (DDT targets)	5	0.8808	0.9312	0.9366	**0.9396**	0.9337	0.9837	0.8750	0.9749	0.9545	0.8198
GFa (geometry feedback)	3	0.8833	0.9326	0.9381	0.9366	0.9396	0.9840	0.8766	0.9765	0.9569	0.8644
GFb (feedback, 1/4)	1	0.8824	0.9321	0.9375	0.9386	0.9365	0.9839	0.8768	0.9758	0.9585	0.8744

Table 6: Pooled test IoU and F1 with seed standard deviation, validation IoU at the selected threshold, selected threshold and training time (GPU-hours per run, NVIDIA L40S).

Model	seeds	test IoU	test F1	val IoU	thr.	GPU-h
Massachusetts						
R0 (ResNeXt-50 U-Net)	0,1,2,3,4	0.6556 ± 0.0018	0.7920 ± 0.0013	0.6470 ± 0.0017	0.40	3.7
R1 (DWT input)	0,1,2	0.6554 ± 0.0024	0.7918 ± 0.0018	0.6470 ± 0.0004	0.40	3.7
R2 (wavelet pool)	0,1,2	0.6545 ± 0.0001	0.7912 ± 0.0001	0.6458 ± 0.0025	0.37	3.8
R3 (R2 + clDice)	0,1,2	0.6553 ± 0.0018	0.7918 ± 0.0013	0.6449 ± 0.0026	0.43	3.9
U-Net (scratch)	0,1,2	0.6632 ± 0.0001	0.7975 ± 0.0001	0.6514 ± 0.0020	0.35	5.0
D-LinkNet	0,1,2	0.6515 ± 0.0014	0.7890 ± 0.0010	0.6391 ± 0.0014	0.38	1.2
GA (aux. centre+width)	0,1,2,3,4	0.6588 ± 0.0008	0.7943 ± 0.0006	0.6495 ± 0.0013	0.43	3.9
GB (drawn mask)	0,1,2	0.6555 ± 0.0004	0.7919 ± 0.0003	0.6462 ± 0.0008	0.35	4.4
GFa (geometry feedback)	0,1,2	0.6591 ± 0.0018	0.7945 ± 0.0013	0.6488 ± 0.0024	0.40	4.2
GFb (feedback, 1/4)	0	0.6584	0.7940	0.6498	0.40	4.4
DeepGlobe						
R0 (ResNeXt-50 U-Net)	0,1,2	0.6771 ± 0.0023	0.8074 ± 0.0016	0.6806 ± 0.0025	0.45	6.1
R1 (DWT input)	0,1,2	0.6785 ± 0.0010	0.8085 ± 0.0007	0.6805 ± 0.0011	0.45	6.1
R2 (wavelet pool)	0,1,2	0.6823 ± 0.0011	0.8111 ± 0.0008	0.6839 ± 0.0016	0.45	6.4
R3 (R2 + clDice)	0,1,2	0.6743 ± 0.0013	0.8054 ± 0.0009	0.6775 ± 0.0002	0.48	6.5
U-Net (scratch)	0,1,2	0.6461 ± 0.0008	0.7850 ± 0.0006	0.6512 ± 0.0005	0.45	8.1
D-LinkNet	0,1,2	0.6655 ± 0.0005	0.7992 ± 0.0004	0.6687 ± 0.0003	0.48	2.1
GA (aux. centre+width)	0,1,2	0.6858 ± 0.0007	0.8136 ± 0.0005	0.6885 ± 0.0005	0.45	6.5
GB (drawn mask)	0,1,2	0.6781 ± 0.0008	0.8082 ± 0.0006	0.6813 ± 0.0003	0.38	7.2
GAc (centre only)	0,1,2	0.6784 ± 0.0008	0.8084 ± 0.0006	0.6808 ± 0.0011	0.45	6.5
GAw (width only)	0,1,2	0.6849 ± 0.0004	0.8130 ± 0.0003	0.6887 ± 0.0009	0.45	6.5
GD (DDT targets)	0,1,2	0.6834 ± 0.0006	0.8119 ± 0.0004	0.6870 ± 0.0008	0.45	6.7
GO (orientation)	0,1,2	0.6812 ± 0.0029	0.8104 ± 0.0020	0.6849 ± 0.0023	0.47	6.6
GFa (geometry feedback)	0,1,2	0.6850 ± 0.0018	0.8131 ± 0.0012	0.6881 ± 0.0003	0.45	7.0
GFb (feedback, 1/4)	0	0.6857	0.8136	0.6890	0.45	7.0
Ottawa						
R0 (ResNeXt-50 U-Net)	0,1,2,3,4	0.8861 ± 0.0048	0.9396 ± 0.0027	0.9015 ± 0.0009	0.48	0.7
GA (aux. centre+width)	0,1,2,3,4	0.8828 ± 0.0036	0.9377 ± 0.0020	0.8922 ± 0.0027	0.52	0.9
GD (DDT targets)	0,1,2,3,4	0.8808 ± 0.0023	0.9366 ± 0.0013	0.8935 ± 0.0027	0.58	0.8
GFa (geometry feedback)	0,1,2	0.8833 ± 0.0024	0.9381 ± 0.0014	0.8981 ± 0.0024	0.43	0.9
GFb (feedback, 1/4)	0	0.8824	0.9375	0.8986	0.65	0.9

11.2 Pre-specified and secondary comparisons

Table 7: All comparisons. n_s = seeds of the first arm. Two intervals as in Section 10.

Comparison	n_s	$\hat{\Delta}$ IoU$_{img}$	two-level 95% CI	seed-paired t 95% CI	Δ pooled	decision	role
Massachusetts							
GB – R0	3	-0.0012	[-0.0050, +0.0027]	[-0.0056, +0.0013]	-0.0002	no detected diff.	primary (geo)
GA – R0	5	+0.0027	[-0.0012, +0.0068]	[-0.0008, +0.0061]	+0.0031	no detected diff.	primary (geo2, 5 seeds)
GFa – R0	3	+0.0032	[-0.0018, +0.0085]	[-0.0031, +0.0077]	+0.0035	no detected diff.	primary (gf_a)
GB – GA	3	-0.0038	[-0.0078, -0.0002]	[-0.0083, -0.0008]	-0.0033	difference	secondary
GFa – GA	3	+0.0006	[-0.0038, +0.0047]	[-0.0062, +0.0059]	+0.0003	no detected diff.	secondary
DeepGlobe							
GB – R0	3	-0.0017	[-0.0059, +0.0028]	[-0.0129, +0.0095]	+0.0011	no detected diff.	primary (geo)
GA – R0	3	+0.0073	[+0.0033, +0.0118]	[+0.0011, +0.0135]	+0.0087	difference	confirmatory (registered 13:29, 15 Oct)
GAc – R0	3	+0.0005	[-0.0033, +0.0051]	[-0.0098, +0.0108]	+0.0013	no detected diff.	secondary
GAw – R0	3	+0.0043	[+0.0002, +0.0088]	[-0.0013, +0.0098]	+0.0078	inconclusive	secondary
GD – R0	3	+0.0027	[-0.0012, +0.0070]	[-0.0048, +0.0101]	+0.0063	no detected diff.	secondary
GO – R0	3	+0.0014	[-0.0041, +0.0068]	[-0.0147, +0.0175]	+0.0041	no detected diff.	secondary
GFa – R0	3	+0.0056	[+0.0012, +0.0103]	[-0.0072, +0.0183]	+0.0080	inconclusive	primary (gf_a)
GB – GA	3	-0.0090	[-0.0118, -0.0062]	[-0.0155, -0.0025]	-0.0077	difference	secondary
GFa – GA	3	-0.0017	[-0.0048, +0.0013]	[-0.0083, +0.0048]	-0.0008	no detected diff.	secondary
GD – GA	3	-0.0046	[-0.0072, -0.0022]	[-0.0068, -0.0025]	-0.0024	difference	secondary
GO – GA	3	-0.0059	[-0.0110, -0.0017]	[-0.0161, +0.0042]	-0.0046	inconclusive	secondary
GAc – GA	3	-0.0068	[-0.0101, -0.0035]	[-0.0111, -0.0025]	-0.0074	difference	secondary
GAw – GA	3	-0.0030	[-0.0057, -0.0004]	[-0.0042, -0.0019]	-0.0009	difference	secondary
GA – GD	3	+0.0046	[+0.0021, +0.0072]	[+0.0025, +0.0068]	+0.0024	difference	primary (geo3)
Ottawa							
GA – R0	5	-0.0029	[-0.0081, +0.0024]	[-0.0105, +0.0047]	-0.0034	no detected diff.	primary (ott)
GD – R0	5	-0.0048	[-0.0101, -0.0001]	[-0.0096, +0.0001]	-0.0054	inconclusive	secondary
GFa – R0	3	-0.0032	[-0.0084, +0.0032]	[-0.0142, +0.0080]	-0.0028	no detected diff.	primary (gf_a)
GFa – GA	3	-0.0002	[-0.0051, +0.0054]	[-0.0094, +0.0119]	+0.0006	no detected diff.	secondary
GD – GA	5	-0.0019	[-0.0062, +0.0019]	[-0.0071, +0.0034]	-0.0020	no detected diff.	secondary

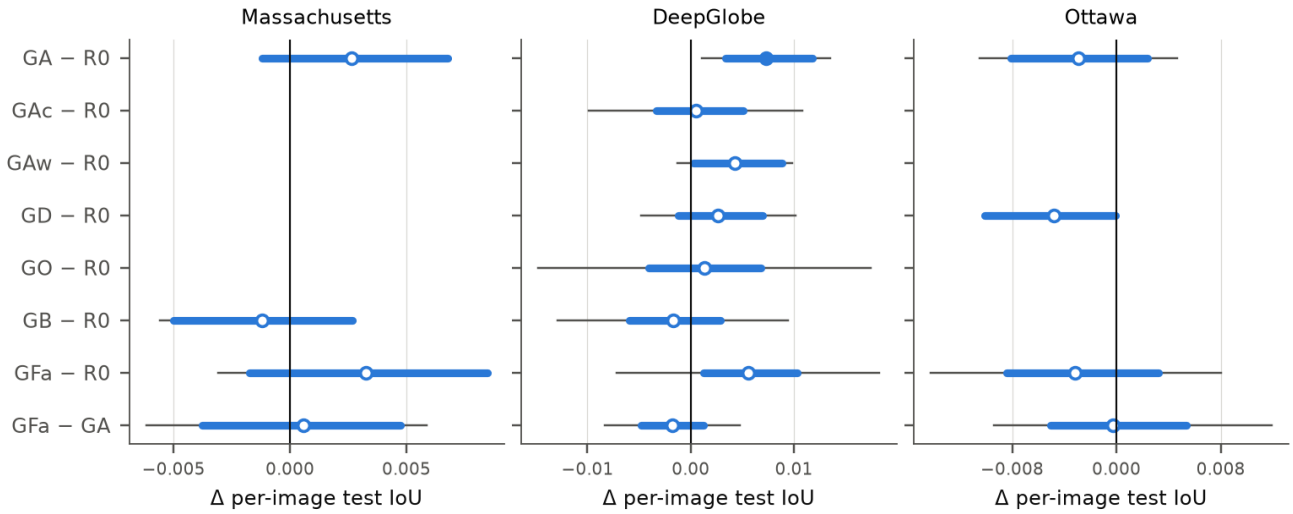


Figure 6: Differences in per-image test IoU. Thick bar: two-level bootstrap 95% CI; thin line: seed-paired t 95% CI; filled marker: "difference" under the decision rule, hollow: not. Only GA − R0 on DeepGlobe meets the rule.

11.3 Attribution of GA's DeepGlobe gain

GAc − R0 +0.0005 (no difference), GAw − R0 +0.0043 (inconclusive: bootstrap $[+0.0002, +0.0088]$, t $[-0.0013, +0.0098]$); GA − GAc +0.0068 and GA − GAw +0.0030 (both differences). By the reading rule fixed in the geo2 plan, neither term alone carries the gain: it needs both, with width contributing

most (pooled 0.6849 vs. GA 0.6858 vs. R0 0.6771).

11.4 Prior geometric targets

On DeepGlobe GA beats DDT targets: GA − GD +0.0046 (bootstrap [+0.0021, +0.0072], t [+0.0025, +0.0068]; per seed +0.0055, +0.0045, +0.0038). GD − R0 +0.0027 and GO − R0 +0.0014: no detected difference. GAw − GD +0.0016: no detected difference — the width encoding alone is not distinguishable from DDT's; the edge over DDT needs the centre term. On Ottawa GD − R0 −0.0048 (inconclusive, R0 ahead).

11.5 Geometry feedback and its selection

Selection on validation IoU (seed 0, threshold-selected): GFa 0.6501 / 0.6879 / 0.9006 (mean 0.7462), GFb 0.6498 / 0.6890 / 0.8986 (mean 0.7458) on Massachusetts / DeepGlobe / Ottawa: GFa selected by 0.0004. Confirmation (seeds 0–2): GFa − R0 +0.0032 / +0.0056 (inconclusive) / −0.0032; GFa − GA +0.0006 / −0.0017 / −0.0002, no detected difference anywhere. On Ottawa GFa recovers part of GA's APLS loss (0.864 vs. 0.839; R0 0.883), a secondary observation on six scenes.

11.6 Error analysis

Error pixels at each run's threshold, split into missed road within 2 px of the label edge (FN edge), missed road deeper inside (FN interior), false road within 2 px of a labelled road (FP near) and farther away (FP far); recall by label width. Means over seeds 0–2.

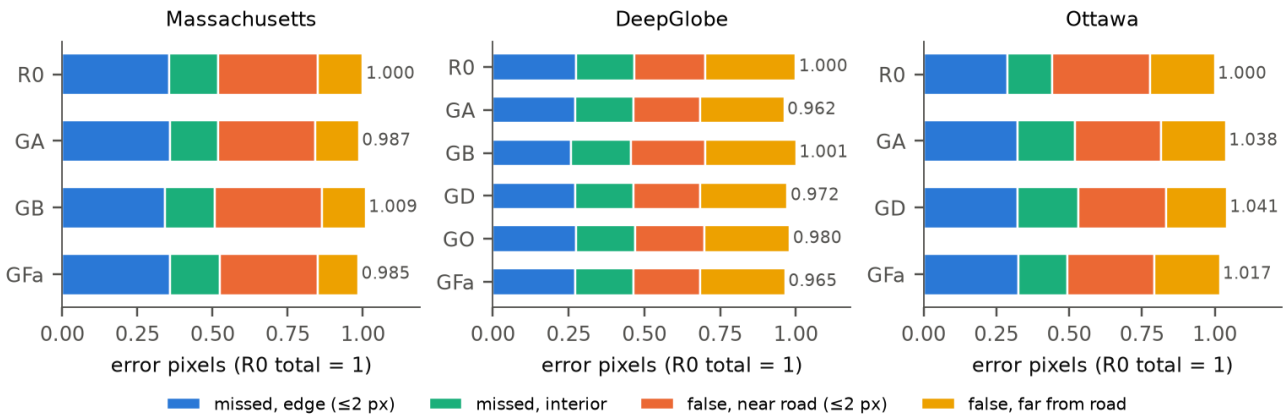


Figure 7: Error composition relative to R0's total (=1). On DeepGlobe GA, GD and GFa remove false positives (orange, yellow); on Ottawa the geometric models add missed-road errors.

Table 8: Share of error pixels by type and recall by label width (px at the dataset's resolution).

	share of errors (%)				recall by label width (px)				
Model	FN edge	FN int.	FP near	FP far	1–4	5–6	7–8	9–12	13–20
Massachusetts									
R0	35.7	16.2	33.2	14.9	0.571	0.789	0.793	0.741	0.468
GA	36.3	16.2	32.8	14.6	0.575	0.789	0.793	0.751	0.483
GB	34.0	16.4	35.3	14.3	0.560	0.791	0.808	0.722	0.453
GFa	36.5	16.6	33.1	13.8	0.568	0.786	0.795	0.760	0.481
DeepGlobe									
R0	27.4	19.4	23.4	29.8	0.555	0.689	0.775	0.826	0.849
GA	28.3	20.0	22.9	28.8	0.564	0.683	0.773	0.829	0.850
GB	25.8	19.6	24.5	30.1	0.559	0.684	0.776	0.834	0.858
GD	28.0	19.8	22.6	29.6	0.548	0.680	0.772	0.825	0.853
GO	28.1	19.7	23.4	28.8	0.555	0.689	0.770	0.824	0.850
GFa	28.2	20.0	22.8	29.0	0.561	0.681	0.772	0.829	0.852
Ottawa									
R0	28.8	15.4	33.7	22.2	0.622	0.547	0.745	0.923	0.955
GA	31.0	19.0	28.5	21.5	0.600	0.542	0.715	0.908	0.945
GD	31.1	20.0	28.8	20.0	0.541	0.497	0.720	0.913	0.943
GFa	32.2	16.4	29.4	22.0	0.605	0.565	0.743	0.912	0.948

On DeepGlobe, relative to R0, GA changes TP by +0.1%, FN by −0.6%, FP-near by −6.1% and FP-far by −7.2%: the gain is precision, with recall at every width unchanged. GB misses 5.8% fewer edge pixels but adds 4.6% near false positives (it draws roads slightly wide). On Ottawa every geometric model loses recall on narrow roads (width 1–8 px).

11.7 Training curves

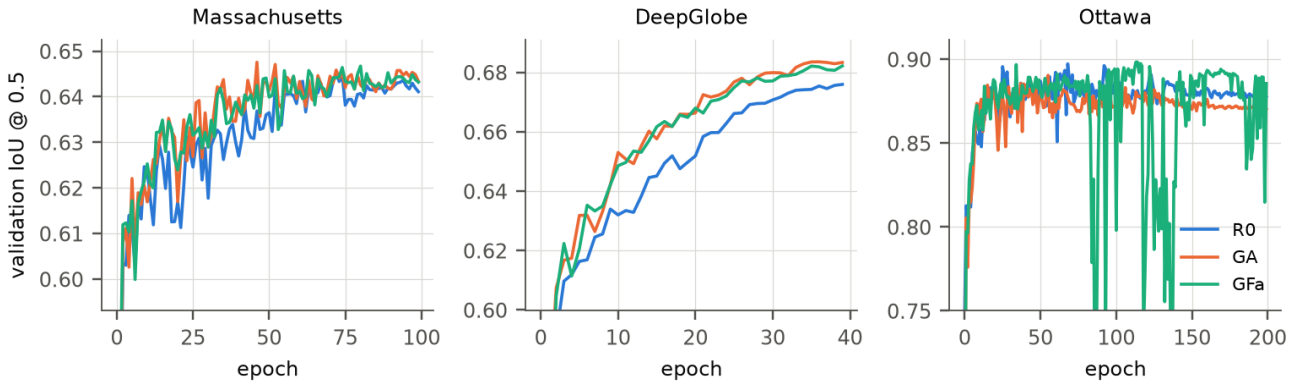


Figure 8: Validation IoU at threshold 0.5 per epoch, seed 0. On DeepGlobe the geometric models lead R0 throughout training. Ottawa's validation set is two scenes, hence the noise (GFa's dips; y-axis clipped at 0.75).

11.8 Qualitative results

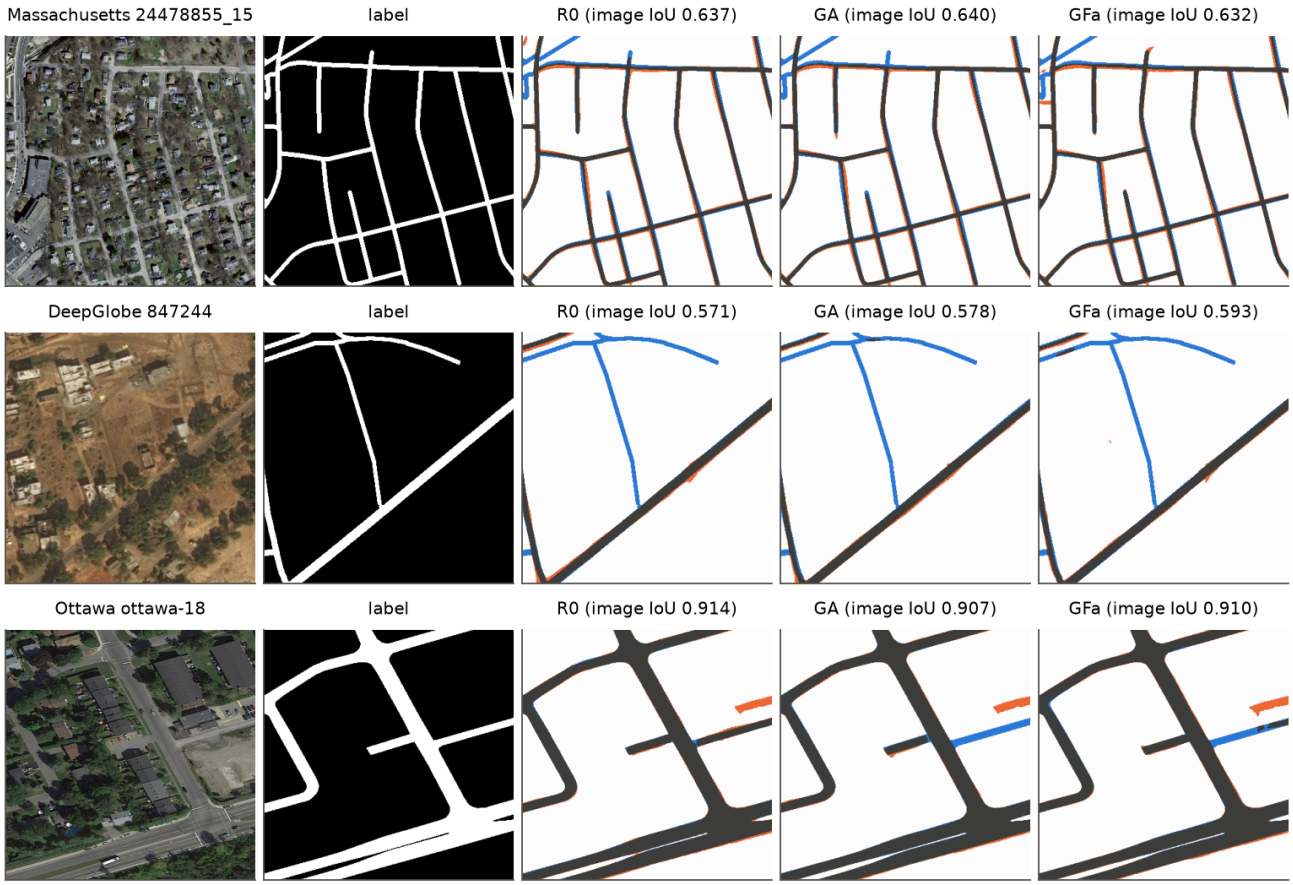


Figure 9: One test image per dataset, chosen by rule: the image whose GA − R0 difference (seed 0) is closest to the dataset's mean difference (typical, not favourable), cropped to the 384^2 window with the most road. Dark grey: correct road; orange: false positive; blue: missed road.

12 Comparison with published results

Table 9: Published results and ours. Values in $[0, 1]$ as printed (percentages converted). *Different splits and protocols: compare within a row group only.*

Method	Protocol	Metric	Massachusetts	DeepGlobe
Massachusetts official split; relaxed breakeven ($\rho = 3$), as reported in [5]				
Mnih-CNN [2] / U-Net / ResUNet [5]	1108/14/49	breakeven	0.8873 / 0.9053 / 0.9187	—
DeepGlobe challenge server				
D-LinkNet [7]	val / test	road IoU	—	0.6466 / 0.6342
NL-LinkNet [8]	val	"mIoU"	—	0.6500
DeepGlobe 4696/1530, full image [24]				
RoadCNN / Orientation+seg. [24]	4696/1530	road IoU (APLS)	—	0.6761 / 0.6721 (0.7312)
Seg-Road protocol [11]				
D-LinkNet / CoANet / Seg-Road-l	Mass 1108/14/49; DG 4696/1530	road IoU	0.6145 / 0.6167 / 0.6838	0.5762 / 0.5911 / 0.6720
URoadNet [12] (images resized to 512^2)				
SegFormer / URoadNet	Mass 1108/14/49; DG 4891/190/1145	IoU	0.6009 / 0.6761	0.6742 / 0.6777
DSFC-Net [13]; RoadFocusNet [14]				
U-Net / D-LinkNet / DSFC-Net	Mass 1108/63; DG 4980/1236	IoU	0.6456 / 0.6443 / 0.6750	0.6506 / 0.6599 / 0.6868
RoadFocusNet	own split	IoU	—	0.7083
This study: Mass 1108/14/49, no-data masked; DG 4358/933/935; stitched full images; mean of seeds				
R0 / GA / GFa	as above	pooled road IoU	0.6556 / 0.6588 / 0.6591	0.6771 / 0.6858 / 0.6850
R0 / GA	as above	F1	0.7920 / 0.7943	0.8074 / 0.8136
R0 / GA	as above	relaxed F1 ($\rho = 3$)	0.9065 / 0.9077	0.8978 / 0.8985
R0 / GA	as above	APLS (ours)	0.8007 / 0.8029	0.7755 / 0.7753

Table 10: Ottawa: RoadNet's road-surface results [25] (Table II; G = global pixel accuracy, C = class-average accuracy, I/U = mean IoU over road and background) on the *same* test scenes 1, 16–20 at native 0.21 m, and ours at 0.42 m (mean of seeds).

Method	Acc. (G)	mIoU (I/U)	Precision	Recall	F1
FCN8s [25]	0.981	0.919	0.917	0.932	0.925
SegNet	0.963	0.853	0.848	0.864	0.856
U-Net	0.973	0.887	0.899	0.885	0.892
CasNet_{++}	0.983	0.929	0.932	0.936	0.934
RoadNet_{++} (best)	0.985	0.934	0.936	0.942	0.939
R0 (ours, 5 seeds)	0.984	0.934	0.933	0.947	0.940
GA (ours, 5 seeds)	0.984	0.932	0.936	0.940	0.938
GFa (ours, 3 seeds)	0.984	0.933	0.937	0.940	0.938

Reading the comparison. Every paper uses its own split and protocol; only rows inside one group are like-for-like. The same architecture (D-LinkNet) is reported at 0.576–0.666 road IoU on DeepGlobe depending on protocol — a spread larger than every architectural gain claimed in the table. Our R0 is level with the strongest DeepGlobe reports under other protocols (0.672–0.687) and, on the Ottawa test scenes, with RoadNet's best model (mIoU 0.934 vs. 0.934, F1 0.940 vs. 0.939), although we evaluate at half resolution. Massachusetts numbers are lower than published ones partly because we exclude no-data pixels and score stitched full images. Seg-Road's printed F1 (0.9143 with road IoU 0.672) is inconsistent with $\text{IoU} = F_1/(2 - F_1)$, so only its road IoU is quoted.

13 Discussion

What helped, and why there. The only confirmed gain is GA on DeepGlobe, made of fewer false positives far from and near roads. DeepGlobe's scenes are rural and peri-urban with many road-like

structures (field boundaries, tracks, dry riverbeds); a width target forces the network to commit to a plausible cross-section, which plausibly suppresses road-like texture without a coherent width. This is an interpretation, not a tested mechanism.

Why it does not transfer. On Massachusetts the labels have a constant width, so a width target carries little information. On Ottawa the labels are precise surfaces, IoU is near 0.88 and precision near 0.93; there is little false-positive mass to remove, and the geometric models lose recall on narrow roads. Ottawa's six test scenes also limit power.

Representation vs. supervision. Drawing the mask from geometry (GB) — the analogue of DDT's reconstruction — did not help on either dataset and was worse than GA; DDT reported large gains from its refinement on CT. Feeding the geometry back (GF) did not add to GA. On these data the geometric information helps, when it helps, as a training signal, not as an output representation.

Against prior targets. On DeepGlobe GA beats DDT's targets; GD and GO do not beat R0. The advantage over DDT needs the centre-line term, since the width encoding alone (GAw) is level with GD.

14 Limitations

- One confirmed positive result, on one dataset; the effect is about +0.7 IoU points per image.
- 3 seeds on DeepGlobe (5 on Massachusetts/Ottawa for R0, GA, GD); the bootstrap can be anti-conservative; Ottawa has 6 test scenes.
- Loss weights (1, 1), radii, crop counts and the recipe were not tuned; one backbone.
- The target design changed twice after inspecting *training* masks (never val/test).
- The DeepGlobe confirmatory test was registered after the Massachusetts result was seen.
- GD and GO reuse our model and recipe, not their papers' architectures (Batra's stacked hourglass and refinement; DDT's 3-D networks and post-hoc refinement).
- Massachusetts scores exclude a custom no-data mask; DeepGlobe uses our own random split from a mirror; Ottawa is used at half resolution with one test scene containing 1.45% black pixels.
- bf16 and non-deterministic kernels: reruns are statistically, not bitwise, reproducible.
- APLS is our implementation (8 px snapping, zero on miss) and not comparable with others'.
- CHN6-CUG was not evaluated (download behind a request form).

15 Reproducibility

Repository `/scratch/hk4488/roadseg` (GitHub `HarshithKoriRaj/roadseg`, private). Branches: `feat/geo-targets` → `feat/geo2-attribution` → `feat/geo3-baselines` → `feat/ott-dataset` → `feat/gf-feedback`. Spec: `docs/superpowers/specs/2026-10-14-geometric-targets-design.md`; ledger (every ruling, job id and result): `docs/superpowers/ledgers/2026-10-14-geometric-targets-ledger.md`. Every run directory records its resolved configuration, seed, code commit, library and CUDA versions, the pretrained-weight hash, the split sizes, the training history, the selected threshold, per-image test scores and test probability maps. This report's numbers and figures are produced by `scripts/analysis/report_paper2.py` from those files.

Table 11: Campaigns (SLURM array and aggregation jobs; code snapshot).

Campaign	Content	Runs	Array / aggregate	Snapshot
geo	GA, GB × Mass, DG × 3 seeds	12	19841566 / 19841568	862387b
geo2	R0, GA seeds 3–4 (Mass); GAc, GAw (DG)	10	20012726 / 20012727	a6cc48b
geo3	GD, GO (DG)	6	20079336 / 20079337	ba1e8e9
ott	R0, GA, GD × 5 seeds (Ottawa)	15	20200093 / 20200094	8cb06c4
gfsel	GFa, GFb seed 0 × 3 datasets	6	20253709 / —	c430651
gfconf_a	GFa seeds 1–2 × 3 datasets	6	20281436 / 20281440	c430651
total		55	≈220 GPU-h (paper 1: 36 runs, ≈170 GPU-h)	

A Paper-1 models in the same table

R1 (Haar DWT at the input with an exact, function-preserving stem), R2 (learned Haar-basis pooling and unpooling), R3 (R2 + clDice), U-Net from scratch and D-LinkNet were trained with the identical pipeline in paper 1 and appear in Table 5 with †. Paper 1's pre-registered result: R1 − R0 +0.0004 (Massachusetts) and +0.0017 (DeepGlobe), no detected difference.

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